Topic A

Multiple choice questions

1 Two cyclists X and Y are separated by a distance of 68 km. At $t = 0$, X starts cycling with speed 8.0 km h^{-1}, and one hour later Y starts cycling with speed 12 km h^{-1}.

What is the displacement of Y when X and Y meet?

A 32 km
B −32 km
C 36 km
D −36 km

2 A car has an initial velocity 20 m s^{-1}. It decelerates at 5.0 m s^{-2}. After which distance will the car stop?

A 4.0 m
B 40 m
C 80 m
D 100 m

3 A small plane takes off at a speed of 30 m s^{-1} after accelerating from rest on a runway for 900 m. What is the acceleration of the plane on the runway?

A 0.50 m s^{-2}
B 1.0 m s^{-2}
C 3.0 m s^{-2}
D 4.0 m s^{-2}

4 An object with initial speed 2.0 m s^{-1} accelerates to a speed of 8.0 m s^{-1} in 4.0 s. What is the distance covered?

A 8.0 m
B 16 m
C 20 m
D 32 m

5 A block accelerating on a straight line with constant acceleration a increases its speed from u to v in a time t. The following are suggested expressions for the displacement of the block during this time:

I $\frac{v^2 - u^2}{2a}$

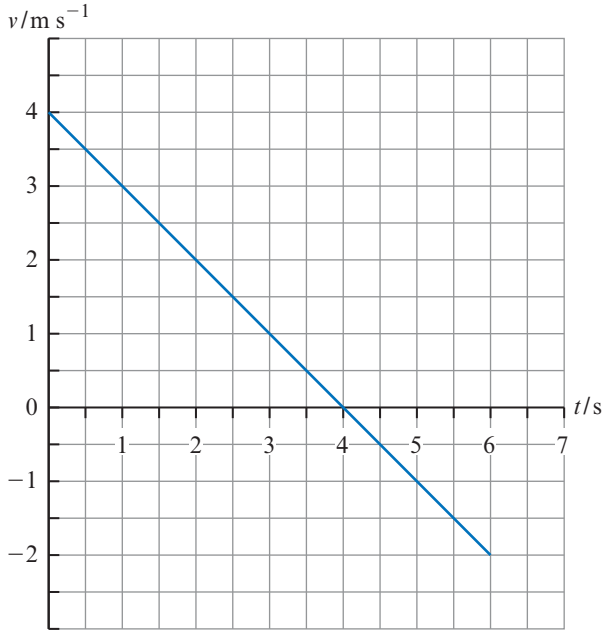
II $ut + \frac{1}{2}at^2$

III $vt - \frac{1}{2}at^2$

Which of the following are correct expressions for the displacement?

A I and II
B I and III
C II and III
D I, II and III.

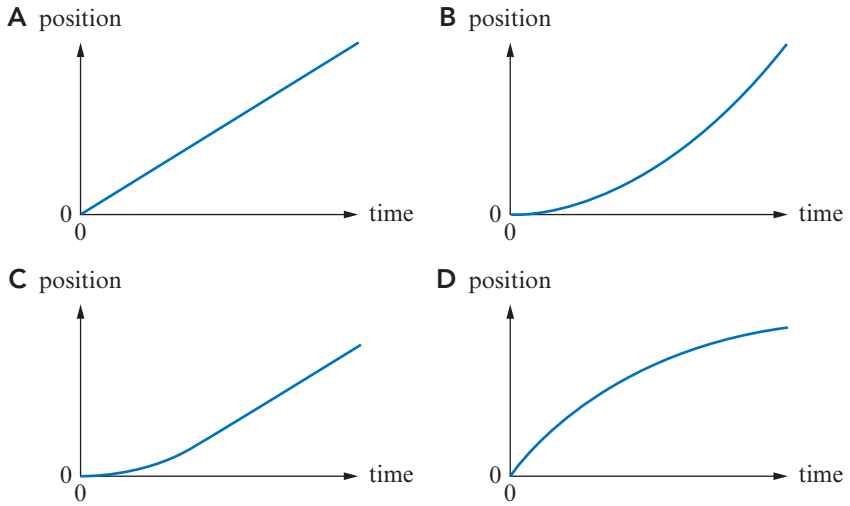
6 The graph shows the variation with time of the velocity of an object.



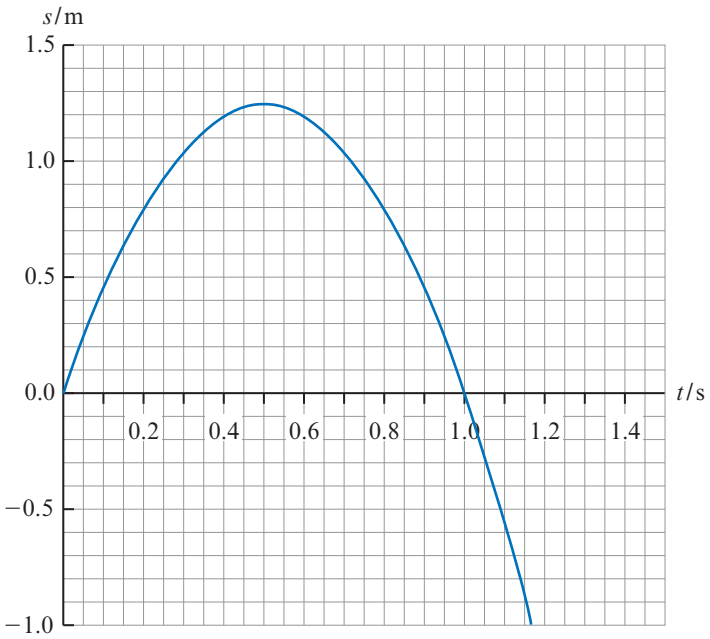
What is the distance and displacement after 6.0 s?

	Distance/m	Displacement/m
A	6	6
B	6	10
C	10	6
D	10	10

7 A ping-pong ball is released from rest and falls vertically. The ball attains terminal speed. Which graph shows the variation of the position of the ball with time?



8 The graph shows how the position of an object varies with time. The shape of the graph is a parabola.



What is correct about the acceleration of the object?

A It is zero.

B It is constant and positive.

C It is constant and negative.

D The acceleration is not zero and not constant.

9 On Earth, a stone takes time T to fall from rest a certain vertical distance and impacts the surface with speed v. What would the time of fall and impact speed be when the stone falls the same vertical distance on a planet where the acceleration of free fall is $\frac{g}{4}$?

	Time to fall	Impact speed
A	$2T$	$\frac{v}{2}$
B	$2T$	$\frac{v}{4}$
C	$4T$	$\frac{v}{2}$
D	$4T$	$\frac{v}{4}$

10 An object thrown vertically upwards reaches a maximum height H. The same object is thrown upwards with the same speed on a planet where the acceleration of free fall is $\frac{g}{3}$. What is the maximum height attained?

A $\frac{H}{3}$

B $\frac{H}{\sqrt{3}}$

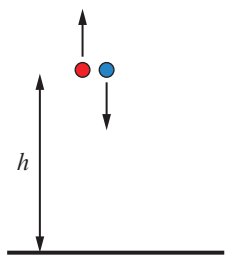
C $H\sqrt{3}$

D $3H$

11 On Earth the magnitude of the acceleration of free fall is $g = 9.8\ \text{m s}^{-2}$. A ball is thrown upwards and after reaching its maximum height falls down again. Taking the upward direction as positive, what is correct about the acceleration of the ball during its motion?

	Ball moving upward	Ball at maximum height	Ball moving downward
A	g	zero	$-g$
B	$-g$	$-g$	$-g$
C	$-g$	zero	g
D	g	g	g

12 Two balls are thrown with the same initial speed of $20\ \text{m s}^{-1}$ from the same height h above ground. One ball is thrown upwards and the other downwards.



What is the time between the arrivals of the balls on the ground?

A 1 s

B 2 s

C 4 s

D Impossible to answer without knowing h.